

Short-Read Isoform Quantification as a GLM

Purpose

Common short-read isoform quantification models can be written in a common generalized linear model (GLM) framework. This formulation also defines the read-isoform alignment conditional probability matrix used in quantification-error analysis and K-value calculation.

Basic Model

Consider a gene with T isoforms and relative abundance vector

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_T)'.$$

There are N independent reads $R_1, \dots, R_N$, each with common read length L . For isoform t , let l_t be the isoform length and

$$\tilde{l}_t = l_t - L + 1$$

be its effective length. Read start sites are assumed to be uniformly distributed over the isoform of origin.

Under this setup, the probability that a read originates from isoform t is

$$\phi_t = \Pr(R_n \in t) = \frac{\theta_t \tilde{l}_t}{\sum_{j=1}^T \theta_j \tilde{l}_j}.$$

Read-Based and Region-Based Views

The read-based likelihood models the probability of observing each read directly. A read contributes according to the isoforms to which it can map:

$$\Pr(\text{observing } R_n) = \sum_{t=1}^T \frac{1}{\tilde{l}_t} \mathbf{1}\{R_n \text{ maps to isoform } t\} \phi_t.$$

The region/equivalence class-based likelihood first partitions a gene into disjoint regions such that reads falling in the same region map to the same set of isoforms (equivalence classes). If region s has effective length $\tilde{u}_s$, the probability that a randomly selected read falls in region s is

$$p_s = \sum_{t=1}^T \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\} \phi_t.$$

Region counts can then be modeled with a multinomial distribution, approximated with independent Poisson variables, or approximated using a normal model for region read proportions.

Main Equivalence Result

In the basic model, read-based and region-based formulations yield equivalent estimators. The key observation is that reads within the same disjoint region have identical mapping compatibility and therefore make the same likelihood contribution. Grouping reads by region turns the read-based likelihood into the multinomial region likelihood up to a constant. The Poisson approximation is also equivalent to these formulations, so the read-based, multinomial, and Poisson likelihoods produce the same maximum-likelihood estimator of θ .

The normal model is not written as the same likelihood, but its mean structure has the same linear form. Thus its estimator approximates the estimators from the read-based, multinomial, and Poisson models.

GLM Formulation

For region read proportions c_s , the normal model uses the identity-link mean equation

$$\mathbb{E}[c_s] = \sum_{t=1}^T \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\} \phi_t, \quad 1 \leq s \leq S.$$

For region read counts n_s , the multinomial and Poisson formulations use

$$\mathbb{E}[n_s] = N \sum_{t=1}^T \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\} \phi_t, \quad 1 \leq s \leq S.$$

Dividing the count equation by N gives the same mathematical relationship as the proportion equation. Both are identity-link GLMs relating observed regional summaries to isoform-origin probabilities.

Conditional Probability Matrix

The normal model can be written as a least-squares problem:

$$\hat{\phi} = \arg \min_{\phi \in \mathbb{R}_+^T} \|\mathbf{c} - A\phi\|_2^2,$$

where $\mathbb{R}_+^T$ denotes the nonnegative orthant. Nonnegativity follows from the interpretation of ϕ_t as an isoform-origin probability or nonnegative isoform signal. A sum-to-one constraint is optional: if $\mathbf{c}$ is normalized and the columns of A have compatible total mass, the least-squares loss identifies the scale of ϕ . The fitted vector can then be normalized post hoc if a probability vector is desired. Here

$$\mathbf{c} = (c_1, c_2, \dots, c_S)'$$

is the vector of region read proportions and

$$A = (a_{st}) \in \mathbb{R}^{S \times T}$$

is the read-isoform alignment conditional probability matrix. Its entries are

$$a_{st} = \Pr(R_n \in s \mid R_n \in t) = \frac{\tilde{u}_s}{\tilde{l}_t} \mathbf{1}\{s \in t\}.$$

Equivalently, the estimation problem is the linear system

$$\mathbf{c} = A\phi.$$

This linear system is the basis for defining the K-value.

Many-Cell Extension and Nuclear-Norm Regularization

For single-cell data, the same gene can be quantified across M cells at once. Let

$$C = [\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_M] \in \mathbb{R}^{S \times M}$$

be the matrix of region read proportions, with one column per cell, and let

$$\Phi = [\phi_1, \phi_2, \dots, \phi_M] \in \mathbb{R}^{T \times M}$$

be the matrix of cell-specific isoform signals. The entries of Φ are constrained to be nonnegative. If the same read-isoform compatibility matrix A is used for all cells, the simultaneous least-squares problem is

$$\hat{\Phi} = \arg \min_{\Phi \geq 0} \frac{1}{2} \|C - A\Phi\|_F^2.$$

If column-normalized isoform proportions are needed, each fitted column can be converted to

$$\tilde{\phi}_m = \frac{\hat{\phi}_m}{\mathbf{1}' \hat{\phi}_m}$$

after optimization, for columns with positive total mass.

Single-cell read coverage is sparse, so both A and C are typically very sparse. To share information across cells, one can either penalize the nuclear norm of the matrix Φ ,

$$\hat{\Phi} = \arg \min_{\Phi \geq 0} \frac{1}{2} \|C - A\Phi\|_F^2 + \lambda \|\Phi\|_*,$$

or constrain it directly:

$$\hat{\Phi} = \arg \min_{\Phi \geq 0, \|\Phi\|_* \leq \tau} \frac{1}{2} \|C - A\Phi\|_F^2.$$

The nuclear norm $\|\Phi\|_*$, the sum of singular values of Φ , favors a low-rank structure in which isoform usage across cells is explained by a small number of shared patterns.

Sparse Objective Evaluation

A scalable implementation should avoid forming the dense residual $C - A\Phi$. The squared loss can be expanded as

$$\|C - A\Phi\|_F^2 = \|C\|_F^2 - 2 \operatorname{tr}(\Phi' A' C) + \operatorname{tr}(\Phi' A' A \Phi).$$

Thus the data enter through

$$B = A' C, \quad G = A' A.$$

Both can be computed with sparse matrix multiplication. The gradient of the smooth loss is

$$\nabla_{\Phi} \frac{1}{2} \|C - A\Phi\|_F^2 = G\Phi - B,$$

which requires multiplication by the small isoform-by-isoform matrix G , rather than by a large dense region-by-cell residual.

Optimization Strategies

There are three practical routes.

- **ADMM with a nuclear-norm split.** Introduce a second matrix Z and solve

$$\min_{\Phi \geq 0, Z} \frac{1}{2} \|C - A\Phi\|_F^2 + \lambda \|Z\|_* \quad \text{subject to} \quad \Phi = Z.$$

The Φ -update is a sparse nonnegative quadratic problem. The corresponding unconstrained normal equation has the form

$$(A'A + \rho I)\Phi = A'C + \rho(Z - U),$$

where U is the scaled dual variable and ρ is the ADMM penalty. Therefore the update can use the sparse products $B = A'C$ and $G = A'A$ instead of forming the dense residual $C - A\Phi$. If T is modest for a gene, factorize $G + \rho I$ once and reuse it for all cells. If T is large, solve the update iteratively using products with sparse A and A' , or with G if G is still sparse enough. The nonnegativity constraint can be handled by projected conjugate gradient, coordinate descent, active-set nonnegative least squares, or an additional proximal projection step.

The Z -update is singular-value soft-thresholding:

$$Z \leftarrow \text{SVT}_{\lambda/\rho}(\Phi + U).$$

This step acts on the isoform-by-cell matrix and does not involve A or C . Thus ADMM can exploit sparsity in the data-fitting step, but the nuclear-norm proximal step may become the bottleneck when M is very large. In that regime, truncated or randomized SVD, cell minibatches, or the factorized formulation below are usually preferable.

- **Frank-Wolfe for a constrained nuclear norm.** For the constrained problem

$$\min_{\Phi \geq 0, \|\Phi\|_* \leq \tau} f(\Phi) = \frac{1}{2} \|C - A\Phi\|_F^2,$$

Frank-Wolfe avoids nuclear-norm proximal steps. At iteration k , compute the gradient

$$D_k = \nabla f(\Phi_k) = G\Phi_k - B.$$

The linear minimization oracle solves

$$S_k = \arg \min_{S \geq 0, \|S\|_* \leq \tau} \langle D_k, S \rangle.$$

Because S is constrained to be nonnegative, only the positive part of the descent matrix matters. Let

$$H_k = (-D_k)_+ = (B - G\Phi_k)_+.$$

If H_k is nonzero, take its leading left and right singular vectors u_k and v_k , which can be chosen nonnegative, and set

$$S_k = \tau u_k v_k'.$$

The update

$$\Phi_{k+1} = (1 - \gamma_k)\Phi_k + \gamma_k S_k$$

preserves both nonnegativity and the nuclear-norm constraint. Line search can use the same sparse sufficient statistics $B = A'C$ and $G = A'A$, since $f(\Phi)$ and directional derivatives are available without forming $C - A\Phi$.

This approach is attractive when a small effective rank is expected: the iterate is an explicit convex combination of rank-one atoms, and each step requires only a leading singular vector computation rather than a full SVD. The main cost is repeatedly computing the top singular vectors of $(B - G\Phi_k)_+$. Randomized or Lanczos methods can exploit fast products with G and the low-rank representation of Φ_k .

- **Low-rank factorization.** Use the variational identity

$$\|\Phi\|_* = \min_{\Phi=UV'} \frac{1}{2} (\|U\|_F^2 + \|V\|_F^2)$$

and optimize a fixed-rank approximation $\Phi \approx UV'$, where $U \in \mathbb{R}^{T \times r}$, $V \in \mathbb{R}^{M \times r}$, and $r \ll \min(T, M)$. This avoids full singular-value decompositions and stores only $O(r(T + M))$ parameters. The loss can be evaluated using B and G :

$$\|C - AUV'\|_F^2 = \|C\|_F^2 - 2 \operatorname{tr}(VU'B) + \operatorname{tr}((U'GU)(V'V)).$$

Nonnegativity can be enforced by projecting UV' onto the nonnegative orthant, by using an ADMM split for the constrained matrix, or by parameterizing Φ through a nonnegative link such as an exponential or softplus transformation. Column normalization can be applied after fitting if isoform proportions are required.

The factorized and Frank-Wolfe approaches are usually the most scalable choices for many cells. Both keep Φ in a compact low-rank representation and use sparse preprocessing for $B = A'C$. ADMM is useful as a direct convex reference implementation for smaller genes or smaller cell batches, but its repeated singular-value thresholding can be more expensive at large scale.

Choosing the Regularization Level

The regularization parameter can be chosen by cross-validation because the many-cell model is a large regression problem. A natural split is over regions, or equivalently over read equivalence classes. For a fold q , split the rows of A and C into training and validation sets:

$$A = \begin{bmatrix} A_{\text{tr}} \\ A_{\text{val}} \end{bmatrix}, \quad C = \begin{bmatrix} C_{\text{tr}} \\ C_{\text{val}} \end{bmatrix}.$$

For each candidate penalty λ , or each candidate nuclear-norm radius τ , fit Φ using only the training rows. For the constrained form, this means solving

$$\hat{\Phi}_\tau^{(q)} = \arg \min_{\Phi \geq 0, \|\Phi\|_* \leq \tau} \frac{1}{2} \|C_{\text{tr}} - A_{\text{tr}}\Phi\|_F^2.$$

Then evaluate prediction error on the held-out rows:

$$\text{CV}_q(\tau) = \left\| C_{\text{val}} - A_{\text{val}} \hat{\Phi}_\tau^{(q)} \right\|_F^2.$$

The selected regularization level minimizes the average validation error across folds. A weighted version can be used if some regions or cells have much higher read depth:

$$\text{CV}_q(\tau) = \left\| W_{\text{val}}^{1/2} \odot (C_{\text{val}} - A_{\text{val}} \hat{\Phi}_\tau^{(q)}) \right\|_F^2,$$

where W_{val} contains inverse-variance or read-depth weights and $\odot$ denotes elementwise multiplication.

Row-wise folds test whether isoform usage inferred from one set of equivalence classes predicts held-out equivalence classes for the same cells. The folds should be stratified so that each training set retains informative regions for the gene; otherwise a fold can remove all evidence distinguishing two isoforms. Holding out individual sparse entries is possible, but holding out rows is usually cleaner because a row corresponds to a biological read compatibility pattern and avoids mixing the same equivalence class between training and validation.

Remarks

- The derivation intentionally starts from a simplified model that only uses isoform of origin and read start site. Full read-based methods can add information such as fragment length, orientation, or GC content.
- In the basic model, equivalence classes coincide with disjoint regions because all reads in a region have the same mapping set and likelihood contribution.
- More complex methods may define equivalence classes differently, but read counts over equivalence classes can still serve as sufficient statistics for isoform abundance estimation.